Yiwei ZHAO | 赵奕炜

Curriculum Vitae

Carnegie Mellon University
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RESEARCH INTERESTS

Cross-Stack Design for Emerging Hardware Systems. Broadly spanning Parallel and Distributed Computing.
Inter-discipline of: Algorithmic Theory + Computer System Design + Computer Architecture and Hardware.

EDUCATION

Carnegie Mellon University, Pittsburgh, Pennsylvania, 2021-present.

- Ph.D. candidate in Computer Engineering.
- Advisor: Prof. Phillip B. Gibbons.

Tsinghua University, Beijing, China, 2017-2021.

- B.E. in Electronic Engineering. Graduated with highest honor.
- Double major in Economics & Finance.

FELLOWSHIPS, HONORS & AWARDS

- Michel and Kathy Doreau Graduate Fellowship (2024 - 2025).
- Lee-Stanziale Ohana Fellowship (2023 - 2024).
- Qualcomm PhD Fellowship Finalist (2024).
- Best Paper Runner-up, VLDB (2023) [6].
- Carnegie Mellon Institute of Technology Dean's Fellow (2021 - 2022).
- Undergraduate with Highest Honor, Tsinghua University & Beijing (2021).
- China National Fellowship for Undergrads (2018 - 2020).

RESEARCH EXPERIENCE

Carnegie Mellon University (Graduate Research Assistant), Pittsburgh, PA. August 2021 – present.

- Thesis Title: Algorithm-System Co-Design of Processing-in-Memory Systems.
- **Foundation of Processing-in-Memory**: Multi-year effort to design provably guaranteed, practically efficient PIM and emerging-hardware systems. Developed index structures [2,5,6] with provably optimal utilization (throughput) and minimal data movement (power) and high real-world performance. Spanning applications across exact nearest-neighbor search in vector databases [1], end-to-end OLTP systems [3], and distributed graph processing [12].

Meta (Research Scientist Intern), Redmond, WA. May 2025 – August 2025; May 2023 – November 2023.

- **AdaVFM** [13] (May 2025 – August 2025): An adaptive on-device inference framework for language-aligned vision foundation models, combining task-aware model scaling, NAS-guided subnet selection, and cloud-assisted multimodal control. State-of-the-art accuracy-efficiency trade-offs for open-vocabulary vision tasks.
- **H4H** [4,10] (May 2023 – November 2023): A two-stage NAS framework that automatically co-optimizes hybrid CNN/ViT models for heterogeneous edge systems combining NPUs and PIM, leveraging architectural heterogeneity and further optimizes hardware and circuits accordingly. State of the art in small on-device vision models.

Tsinghua University (Undergraduate Researcher), Beijing, China, September 2017 – June 2021.

- Thesis: Error Tolerant Designs for ReRAM based Compute-In-Memory Accelerators.

PUBLICATIONS

Full Publications

[1] **Yiwei Zhao**, Hongbo Kang, Ziyang Men, Yan Gu, Guy E. Blelloch, Laxman Dhulipala, Charles McGuffey, and Phillip B. Gibbons. 2026. "**PIM-zd-tree: A Fast Space-Partitioning Index Leveraging Processing-in-Memory**". In Proceedings of the 31st ACM SIGPLAN Annual Symposium on Principles and Practice of Parallel Programming (**PPoPP** '26). Association for Computing Machinery, New York, NY, USA. doi:10.1145/3774934.3786411.

[2] **Yiwei Zhao**, Hongbo Kang, Yan Gu, Guy E. Blelloch, Laxman Dhulipala, Charles McGuffey, and Phillip B. Gibbons. 2025. "**Optimal Batch-Dynamic kd-trees for Processing-In-Memory with Applications**". In Proceedings of the 37th ACM Symposium on Parallelism in Algorithms and Architectures (**SPAA** '25). Association for Computing Machinery, New York, NY, USA, 350–366. doi:10.1145/3694906.3743318.

[3] Hyoungjoo Kim, **Yiwei Zhao**, Andrew Pavlo, and Phillip B. Gibbons. 2025. "**No Cap, This Memory Slaps: Breaking Through the Memory Wall of Transactional Database Systems with Processing-in-Memory**". In Proceedings of the VLDB Endowment (**PVLDB**), 18(11): 4241-4254, July 2025. doi:10.14778/3749646.3749690.

[4] **Yiwei Zhao**, Jinhui Chen, Sai Qian Zhang, Syed Shakib Sarwar, Kleber Hugo Stangherlin, Jorge Tomas Gomez, Jae Sun Seo, Barbara De Salvo, Chiao Liu, Phillip B. Gibbons, Ziyun Li. 2025. "**H4H: Hybrid Convolution-Transformer Architecture Search for NPU-CIM Heterogeneous Systems for AR/VR Applications**". In Proceedings of the 30th Asia and South Pacific Design Automation Conference (**ASPDAC** '25). Association for Computing Machinery, New York, NY, USA, 1133–1141. doi:10.1145/3658617.3697627.

[5] Hongbo Kang, **Yiwei Zhao**, Guy E. Blelloch, Laxman Dhulipala, Yan Gu, Charles McGuffey, and Phillip B. Gibbons. 2023. "**PIM-trie: A Skew-Resistant Trie for Processing-in-Memory**". In Proceedings of the 35th ACM Symposium on Parallelism in Algorithms and Architectures (**SPAA** '23). Association for Computing Machinery, New York, NY, USA, pp. 1–14. doi:10.1145/3558481.3591070.

[6] Hongbo Kang, **Yiwei Zhao**, Guy E. Blelloch, Laxman Dhulipala, Yan Gu, Charles McGuffey, and Phillip B. Gibbons. 2022. "**PIM-tree: A Skew-resistant Index for Processing-in-Memory**". In Proceedings of the VLDB Endowment (**PVLDB**), 16(4): 946-958, December 2022. doi:10.14778/3574245.3574275. arXiv:2211.10516. ***Best Research Paper Runner-up in VLDB 2023***.

[7] Zeyan Li, Junjie Chen, Yihao Chen, Chengyang Luo, **Yiwei Zhao**, Yongqian Sun, Kaixin Sui, Xiping Wang, Dapeng Liu, Xing Jin, Qi Wang, and Dan Pei. 2023. "**Generic and Robust Root Cause Localization for Multi-Dimensional Data in Online Service Systems**". In Journal of Systems and Software (**JSS**), Vol. 203, (2023), 111748. doi:10.1016/j.jss.2023.111748.

[8] Zeyan Li, Chengyang Luo, **Yiwei Zhao**, Yongqian Sun, Kaixin Sui, Xiping Wang, Dapeng Liu, Xing Jin, Qi Wang, and Dan Pei. 2019. "**Generic and Robust Localization of Multi-Dimensional Root Cause**". In the 30th International Symposium on Software Reliability Engineering (**ISSRE** '19). Oct. 28-31, 2019, Berlin. doi:10.1109/ISSRE.2019.00015.

Short Publications, Posters & Workshops

[9] **Yiwei Zhao**, Jinhui Chen, Sai Qian Zhang, Syed Shakib Sarwar, Kleber Hugo Stangherlin, Jorge Tomas Gomez, Jae Sun Seo, Phillip B. Gibbons, Barbara De Salvo, Chiao Liu, Ziyun Li. 2025. "**H4H: Hybrid Convolution-Transformer Architecture Search for NPU-CIM Heterogeneous Systems for AR/VR Applications (Abstract)**". In Proceedings of the 3rd Highlights of Parallel Computing Workshop (**HOPC** '25), July 28, 2025, Portaland, OR, USA. doi:10.1145/3746238.3746241.

[10] **Yiwei Zhao**, Ziyun Li, Win-San Khwa, Xiaoyu Sun, Sai Qian Zhang, Syed Shakib Sarwar, Kleber Hugo Stangherlin, Yi-Lun Lu, Jorge Tomas Gomez, Jae Sun Seo, Phillip B. Gibbons, Barbara De Salvo, Chiao Liu. 2024. "**Neural Architecture Search of Hybrid Models for NPU-CIM Heterogeneous AR/VR Devices**". In 61th ACM/IEEE Design Automation Conference (**DAC** '24), Poster Session, San Francisco, CA, USA, 2024. arXiv:2410.08326.

[11] Hongbo Kang, **Yiwei Zhao**, Guy E. Blelloch, Laxman Dhulipala, Yan Gu, Charles McGuffey, and Phillip B. Gibbons. 2023. "**PIM-tree: A Skew-resistant Index for Processing-in-Memory (Abstract)**". In Proceedings of the 2023 ACM Workshop on Highlights of Parallel Computing (**HOPC** '23), June 16, 2023, Orlando, FL, USA. doi:10.1145/3597635.3598029.

Preprints & Works Under Review

[12] **Yiwei Zhao**, Qiushi Lin, Hongbo Kang, Guy E. Blelloch, Laxman Dhulipala, Yan Gu, Charles McGuffey, and Phillip B. Gibbons. "**Efficient Task-Data Orchestration for Distributed Systems with Application to Graph Processing**". arXiv:2511.11843. Under review for SC '26.

[13] **Yiwei Zhao**, Yi Zheng, Huapeng Su, Jieyu Lin, Stefano Ambrogio, Cijo Jose, Michael Ramamonjisoa, Patrick Labatut, Barbara De Salvo, Chiao Liu, Phillip B. Gibbons, Ziyun Li. "**AdaVFM: Adaptive Vision Foundation Models for Edge Intelligence via LLM-Guided Execution**". arXiv:2604.15622. Under review for ECCV '26.

[14] Hongbo Kang, **Yiwei Zhao**, Kunal Agrawal, Yongwei Wu, and Phillip B. Gibbons. "**Non-Clairvoyant Scheduling for Processing-in-Memory**". Under review for SPAA '26.

[15] Hyoungjoo Kim, **Yiwei Zhao**, Rutwik Pandit, Matteo Interlandi, Carlo Curino, Andrew Pavlo, and Phillip B. Gibbons. "**These Indexes Hit: Practical GPU Indexes for Concurrent Dynamic Updates and Variable-Length Keys**". Under review for VLDB '27.

[16] Hongbo Kang, Xiangyun Ding, **Yiwei Zhao**, Yingdi Shan, Guy E. Blelloch, Laxman Dhulipala, Yan Gu, Charles McGuffey, Mingxing Zhang, Yongwei Wu, and Phillip B. Gibbons. "**FH-index: A Finger-Hinted Learned Index**". Under review for VLDB '27.

[17] Mohammad Bakhshalipour, Valerie Choung, Rutwik Pandit, Sean Hyacinthe, **Yiwei Zhao**, and Phillip B. Gibbons. "**Benchmarking Memory Systems with a Diverse Zoo: Where Does Time Go?**". Under review for ASPLOS '27.

INVITED TALKS

(All conference paper presentations are excluded due to space limit; please refer to the *Publications* for the complete list.)

- **October 2025, UC Berkeley, Simons Institute for the Theory of Computing. In Algorithmic Foundations for Emerging Computing Technologies**: Optimal Semi-Balanced Trees for Processing-in-Memory.
- **October 2025. UC Berkeley. In Programming System Seminar**: Building HPC Systems for Near-Data-Processing: Theory and Practice.
- **October 2025, Stanford University. In Software Research Seminar**: Building Programming Systems for Near-Data-Processing: Theory and Practice.
- **October 2024, CMU. In Parallel Data Lab (PDL) Retreat**: Fast and Principled Techniques for Heterogeneous Compute and Memory.
- **November 2023, CMU**. **In Parallel Data Lab (PDL) Retreat**: System Design on Processing-In-Memory: Starting from Database Systems.
- **November 2022, CMU**. **In Parallel Data Lab (PDL) Retreat**: PIM-tree: A Theoretically and Practically Efficient Index for Processing-In-Memory.

TEACHING EXPERIENCES

- **18-751 Applied Stochastic Processes, with Applications to AI/ML**: Teaching Assistant, Fall 2024, CMU.
- **18-742 Computer Architecture and Systems**: Teaching Assistant, Spring 2024, CMU.

SERVICES

- **38th ACM Symposium on Parallelism in Algorithms and Architectures** (**SPAA'26**): Shadow Program Committee.
- **SIAM Symposium on Algorithm Engineering and Experiments** (**ALENEX26**): Artifact Evaluation Committee.
- **37th ACM Symposium on Parallelism in Algorithms and Architectures** (**SPAA'25**): Junior Program Committee.
- **2025 International Conference on Parallel Architectures and Compilation Techniques** (**PACT'25**): External Review Committee.
- **Student Council for Departmental Faculty Hiring**: Chair, January 2024 – June 2025, CMU.
- **Student Council for Departmental Faculty Hiring**: Member, January 2023 – January 2024, CMU.

REFERENCES

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